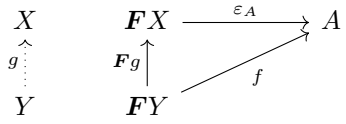
Chapter 5

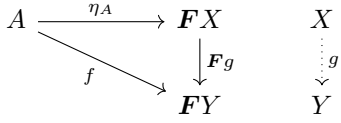
Universal Arrows

5.1 Universal Arrows

Definition 5.1.1 (Universal arrows[1])**.** Given any functor $\boldsymbol{F} : \mathcal{X} \longrightarrow \mathcal{A}$ and object A of $\mathcal{A}$, a *universal arrow from* $\boldsymbol{F}$ *to* A consists of an object X of $\mathcal{X}$ and a morphism $\varepsilon_A : \boldsymbol{F}X \longrightarrow A$ such that given any object Y of $\mathcal{X}$ and morphism $f : \boldsymbol{F}Y \longrightarrow A$ there is a unique morphism $g : Y \longrightarrow X$ of $\mathcal{X}$ that makes the following triangle commute:



Dually, a *universal arrow from* A *to* $\boldsymbol{F}$ consists of an object X of $\mathcal{X}$ and a morphism $\eta_A : A \longrightarrow \boldsymbol{F}X$ such that given any object Y of $\mathcal{X}$ and morphism $f : A \longrightarrow \boldsymbol{F}Y$ there is a unique morphism $g : X \longrightarrow Y$ of $\mathcal{X}$ that makes the following triangle commute:



Example 5.1.2. For any objects A and B of a category $\mathcal{C}$ with all finite products, the pair B^A and $ev_{A,B}$ is universal from the functor $_ \times A$ to B. In this case, the adjuncts are called transposes.

Theorem 5.1.3. [2]If X, ε_A is universal from $\boldsymbol{F}$ to A and there is an isomorphism $i : Y \longrightarrow X$ then $Y, \varepsilon_A \circ \boldsymbol{F}i$ is also universal from $\boldsymbol{F}$ to A.
Conversely, if X, ε_A is universal from $\boldsymbol{F}$ to A and Y, ε'_A is too, and then the unique morphism $i : Y \longrightarrow X$ with $\varepsilon'_A = \varepsilon_A \circ \boldsymbol{F}i$ is iso.

Exercise 5.1.4. Prove Theorem 5.1.3.

5.2 Limits/Colimits as Universal Arrows

Definition 5.2.1 (Diagram of Shape $\mathbb{I}$)**.** A *diagrams of shape* $\mathbb{I}$ are functors $\mathbb{I} \longrightarrow \mathcal{C}$ in which $\mathbb{I}$ is small.

$\mathbb{I}$ is called an *index category*, which is often finite.

Definition 5.2.2 (Constant Diagram[3])**.** For an object A of $\mathcal{C}$, the *constant diagram* on A is the constant functor $\Delta A : \mathbb{I} \longrightarrow \mathcal{C}$ which sends all objects of $\mathbb{I}$ to A and all morphisms of $\mathbb{I}$ to id_A.

Definition 5.2.3 (Diagonal Functor)**.** The *diagonal functor* $\Delta : \mathcal{C} \longrightarrow \mathcal{C}^{\mathbb{I}}$ is the functor which sends an object A of $\mathcal{C}$ to the constant diagram ΔA on A and a morphism $f : A \longrightarrow B$ of $\mathcal{C}$ to the natural transformation $\Delta f : \Delta A \dot{\longrightarrow} \Delta B$ with components $(\Delta f)_I \stackrel{def}{\equiv} f$ where I is an object of $\mathbb{I}$.

[1]See McLarty (1992, p.88).
[2]See McLarty (1992, p.88, Theorem 10.1).
[3]Crole (1993, p.91)

Definition 5.2.4 (Limits and Colimits)**.**

- A *limit* for a diagram $\boldsymbol{D} : \mathbb{I} \longrightarrow \mathcal{C}$ is a universal arrow $\langle r, v \rangle$ from Δ to $\boldsymbol{D}$. It consists of an object $r \in ob\mathcal{C}$, represented as $r = \varprojlim D$ or $LimD$, and a natural transformation $v : \Delta r \xrightarrow{\cdot} D$, which is universal among natural transformations $\tau : \Delta c \xrightarrow{\cdot} D$ for objects $c \in ob\mathcal{C}$.
- A *colimit* for a diagram $\boldsymbol{D}$ is a universal arrow $\langle r, u \rangle$ from $\boldsymbol{D}$ to Δ. It consists of an object $r \in ob\mathcal{C}$, represented as $r = \varinjlim D$ or $ColimD$, and natural transformation $u : D \xrightarrow{\cdot} \Delta r$ which is universal among natural transformations $\tau : D \xrightarrow{\cdot} \Delta c$ for objects $c \in ob\mathcal{C}$.

Example 5.2.5. Let $\mathbb{I}$ be the empty category. Then the limit and the colimit for shape $\mathbb{I}$ are a terminal and initial objects.

Example 5.2.6. Let $\mathbb{I}$ be the small discrete category (namely, a category for which the only morphisms are identities). Then the limit and the colimit of shape $\mathbb{I}$ amount to products and coproducts.

Example 5.2.7. Let $\mathbb{I}$ be the category

$$0 \rightrightarrows 1$$

which has just two objects and two parallel non-identity morphisms. The limit and colimit are eqalizer and coequalizer.

Example 5.2.8. Let $\mathbb{I}$ be the category:

$$\begin{array}{ccc} & & 1 \\ & & \downarrow \\ 0 & \longrightarrow & 2 \end{array}$$

The limit and colimit for the diagram are a pullback and a pushout.

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